

Self-Similarity of the Fake-Degree Schur Sum at a Root of Unity

Clio

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Abstract

For $n \geq 1$, $e \geq 2$, and ζ_e a primitive e -th root of unity, define

$$q_e^{(n)} := \sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) s_\lambda \in \Lambda_n \otimes \mathbb{Z}[\zeta_e],$$

where $\tilde{f}_\lambda(t)$ is the fake-degree polynomial. We prove the self-similarity identity

$$q_e^{(n)} = p_e^{\lfloor n/e \rfloor} \cdot q_e^{(n \bmod e)}$$

in the algebra of symmetric functions over $\mathbb{Z}[\zeta_e]$. In particular, the residue factor $F_e := q_e^{(n)} / p_e^{\lfloor n/e \rfloor}$ (a well-defined symmetric function of degree $n \bmod e$, by our earlier support theorem) depends on n only through $r_0 := n \bmod e$, and its Schur expansion is given by the “small” partition-of- r_0 analogue:

$$F_e = \sum_{\lambda \vdash r_0} \tilde{f}_\lambda(\zeta_e) s_\lambda.$$

The proof is a one-line consequence of the closed-form coefficient formula derived in [1], using the classical identity $\prod_{r=1}^{e-1} (1 - \zeta_e^r) = e$.

1 Statement and Context

Fix $e \geq 2$ and let $\zeta_e := e^{2\pi i/e}$. For $n \geq 0$ set

$$q_e^{(n)} := \sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) s_\lambda,$$

where the sum runs over partitions of n and $\tilde{f}_\lambda(t) := \sum_{T \in \text{SYT}(\lambda)} t^{\text{maj}(T)}$ is the fake degree polynomial of λ (equivalently, the graded multiplicity of the irreducible S_n -module V_λ in the coinvariant algebra of S_n). By convention $q_e^{(0)} = 1$.

In [1] we proved the *support theorem*: writing $q_e^{(n)} = \sum_\mu c_\mu^{(n)} p_\mu$ in the power-sum basis,

$$c_\mu^{(n)} \neq 0 \iff \mu = (e^k, \nu), \quad k := \lfloor n/e \rfloor, \quad \nu \vdash r_0 := n \bmod e,$$

and moreover, with the notation $\alpha_r := 1 - \zeta_e^r$ and $b_r(n) := k + \mathbf{1}[r \leq r_0]$,

$$c_{(e^k, \nu)}^{(n)} = \frac{1}{e^k z_\nu} \prod_{r=1}^{e-1} \alpha_r^{b_r(n) - m_r(\nu)}, \tag{1}$$

where $m_r(\nu)$ is the multiplicity of r in ν and $z_\nu = \prod_i i^{m_i(\nu)} m_i(\nu)!$.

Since the parts of ν are $\leq r_0 < e$, $p_{(e^k, \nu)} = p_e^k p_\nu$, so the support theorem factorises $q_e^{(n)}$ as

$$q_e^{(n)} = p_e^k F_e^{(n)}, \quad F_e^{(n)} := \sum_{\nu \vdash r_0} c_{(e^k, \nu)}^{(n)} p_\nu \in \Lambda_{r_0} \otimes \mathbb{Z}[\zeta_e].$$

The main question left open by [1] was the structure of the “residue factor” $F_e^{(n)}$.

Theorem 1 (Self-similarity). *For every $n \geq 0$ and $e \geq 2$, with $k = \lfloor n/e \rfloor$ and $r_0 = n \bmod e$,*

$$q_e^{(n)} = p_e^k \cdot q_e^{(r_0)}.$$

Equivalently, $F_e^{(n)} = q_e^{(r_0)}$; in particular the residue factor depends on n only through r_0 , and its Schur expansion is

$$F_e^{(n)} = \sum_{\lambda \vdash r_0} \tilde{f}_\lambda(\zeta_e) s_\lambda.$$

2 Proof

The proof is a direct calculation at the level of power-sum coefficients, using the closed form (1) and the classical cyclotomic identity below.

Lemma 2. $\prod_{r=1}^{e-1} (1 - \zeta_e^r) = e.$

Proof. Setting $x = 1$ in the factorisation $x^e - 1 = (x - 1) \prod_{r=1}^{e-1} (x - \zeta_e^r)$ and taking the limit,

$$\prod_{r=1}^{e-1} (1 - \zeta_e^r) = \lim_{x \rightarrow 1} \frac{x^e - 1}{x - 1} = e. \quad \square$$

Proof of Theorem 1. If $r_0 = 0$: then $b_r(n) = k$ for all $r \in [1, e - 1]$, and ν ranges only over the empty partition, so $F_e^{(n)}$ is the constant symmetric function

$$F_e^{(n)} = \frac{1}{e^k} \prod_{r=1}^{e-1} \alpha_r^k = \frac{1}{e^k} e^k = 1$$

by Lemma 2, and $q_e^{(0)} = 1$ by convention, so the theorem holds.

Now assume $r_0 \geq 1$. Apply the same closed form (1) at parameter r_0 (in place of n): then $\lfloor r_0/e \rfloor = 0$ (since $0 < r_0 < e$), and $b_r(r_0) = 0 + \mathbf{1}[r \leq r_0] = \mathbf{1}[r \leq r_0]$. So for every $\nu \vdash r_0$,

$$c_\nu^{(r_0)} = \frac{1}{z_\nu} \prod_{r=1}^{e-1} \alpha_r^{\mathbf{1}[r \leq r_0] - m_r(\nu)}. \quad (2)$$

Meanwhile from (1) at parameter n , with $b_r(n) = k + \mathbf{1}[r \leq r_0]$,

$$\begin{aligned} c_{(e^k, \nu)}^{(n)} &= \frac{1}{e^k z_\nu} \prod_{r=1}^{e-1} \alpha_r^{k + \mathbf{1}[r \leq r_0] - m_r(\nu)} \\ &= \frac{1}{e^k z_\nu} \left(\prod_{r=1}^{e-1} \alpha_r \right)^k \prod_{r=1}^{e-1} \alpha_r^{\mathbf{1}[r \leq r_0] - m_r(\nu)} \\ &= \frac{1}{e^k z_\nu} e^k \prod_{r=1}^{e-1} \alpha_r^{\mathbf{1}[r \leq r_0] - m_r(\nu)} && \text{(by Lemma 2)} \\ &= c_\nu^{(r_0)}. && \text{(by (2))} \end{aligned}$$

Therefore

$$\begin{aligned}
q_e^{(n)} &= \sum_{\nu \vdash r_0} c_{(e^k, \nu)}^{(n)} p_{(e^k, \nu)} && \text{(support theorem)} \\
&= \sum_{\nu \vdash r_0} c_{(e^k, \nu)}^{(n)} p_e^k p_\nu \\
&= p_e^k \sum_{\nu \vdash r_0} c_\nu^{(r_0)} p_\nu && \text{(by (by (2)))} \\
&= p_e^k q_e^{(r_0)}. && \text{(support theorem at } r_0: \text{ all } \nu \vdash r_0 \text{ have parts } < e)
\end{aligned}$$

The last step uses that when $r_0 < e$ the support theorem at parameter r_0 has $\lfloor r_0/e \rfloor = 0$, so *all* partitions $\nu \vdash r_0$ contribute (their parts are automatically not divisible by e).

The Schur-basis statement follows: $F_e^{(n)} = q_e^{(r_0)} = \sum_{\lambda \vdash r_0} \tilde{f}_\lambda(\zeta_e) s_\lambda$ by definition of $q_e^{(r_0)}$. $\square$

3 Consequences

Corollary 3 (*n*-independence). *The residue factor $F_e^{(n)} := q_e^{(n)}/p_e^{\lfloor n/e \rfloor}$ depends on (n, e) only through $(e, n \bmod e)$.*

Proof. Theorem 1 identifies $F_e^{(n)}$ with $q_e^{(n \bmod e)}$, which manifestly depends only on $(e, n \bmod e)$. $\square$

Corollary 4 (Multiplicative recursion). *The sequence $(q_e^{(n)})_{n \geq 0}$ satisfies*

$$q_e^{(n)} = p_e \cdot q_e^{(n-e)} \quad \text{for all } n \geq e.$$

Equivalently, $q_e^{(qe+r)} = p_e^q q_e^{(r)}$ for $q \geq 0$ and $0 \leq r < e$.

Proof. Direct from Theorem 1: $q_e^{(n)} = p_e^{\lfloor n/e \rfloor} q_e^{(n \bmod e)}$ and $q_e^{(n-e)} = p_e^{\lfloor n/e \rfloor - 1} q_e^{(n \bmod e)}$ have the same second factor. $\square$

Corollary 5 (Schur closed form for F_e). *For every n with $r_0 = n \bmod e$,*

$$F_e^{(n)} = \sum_{\lambda \vdash r_0} \tilde{f}_\lambda(\zeta_e) s_\lambda.$$

In particular the Schur coefficient of s_λ in $F_e^{(n)}$ equals $\tilde{f}_\lambda(\zeta_e)$: the fake degree of λ evaluated at ζ_e .

Corollary 5 explains the empirical pattern observed in the residue-factor probes of 2026-08-06:

- **Boundary shapes have pure-root-of-unity coefficients.** For $\lambda = (r_0)$: $\tilde{f}_{(r_0)}(t) = 1$, so the coefficient of $s_{(r_0)}$ is 1.

For $\lambda = (1^{r_0})$: $\text{SYT}(1^{r_0})$ has a unique tableau T with $\text{maj}(T) = \binom{r_0}{2}$, so $\tilde{f}_{(1^{r_0})}(t) = t^{r_0(r_0-1)/2}$, and the coefficient of $s_{(1^{r_0})}$ is $\zeta_e^{r_0(r_0-1)/2}$.

- **Interior shapes have cyclotomic-unit magnitudes.** For $\lambda = (2, 1) \vdash 3$: $\text{SYT}(2, 1) = \{[1\ 2; 3], [1\ 3; 2]\}$ with major indices $\{2, 1\}$ respectively, so $\tilde{f}_{(2, 1)}(t) = t + t^2 = t(1 + t)$. At ζ_e : $\zeta_e(1 + \zeta_e)$, with $|1 + \zeta_e| = 2 \cos(\pi/e)$. For $e = 5$ this gives the golden ratio $\varphi = 2 \cos(\pi/5)$; for $e = 7$ it gives $2 \cos(\pi/7)$; for $e = 4$ it gives $\sqrt{2}$. Each of these values matches the empirical probe.

Remark 6 (Semantic content of the theorem). Theorem 1 says that at the point $t = \zeta_e$ on the unit circle, the “mass” of the fake-degree Schur sum concentrates entirely on the coinvariant-algebra shape (e^k, ν) dictated by the arithmetic of n modulo e ; and, more surprisingly, the constant of proportionality is itself the same sum truncated to partitions of $r_0 = n \bmod e$. There is nothing new happening as n increases past the “fundamental domain” $\{0, 1, \dots, e-1\}$: the sequence $q_e^{(n)}$ is a p_e -linear extension of its first e terms.

Remark 7 (The residue factor is really a q_e in miniature). Corollary 5 shows that $F_e^{(n)}$ is not merely a small polynomial in the roots of unity ζ_e^j ; it has the same intrinsic character as q_e itself, only at parameter $r_0 < e$. The recursion in Corollary 4 then makes the entire family $\{q_e^{(n)}\}_{n \geq 0}$ into a $\mathbb{Z}[\zeta_e]$ -lattice generated by the e “basic” sums $q_e^{(0)}, q_e^{(1)}, \dots, q_e^{(e-1)}$ under multiplication by p_e .

Remark 8 (Comparison with Krařkiewicz–Weyman). Krařkiewicz–Weyman’s theorem [3] gives $\tilde{f}_\lambda(\zeta_e) = 0$ whenever λ has non-empty e -core (in the generic Springer-regularity case). At first glance this suggests the Schur expansion of $q_e^{(r_0)}$ should be supported on partitions with empty e -core — but for $\lambda \vdash r_0 < e$, the e -core condition is automatic (every partition of size $< e$ has e -core equal to itself, hence non-empty!) so KW does *not* produce vanishing here. Instead all $\lambda \vdash r_0$ contribute with generically nonzero coefficient $\tilde{f}_\lambda(\zeta_e)$, which is a cyclotomic integer of degree $\leq n(\lambda) \leq \binom{r_0}{2} < \binom{e}{2}$. The KW vanishing is invisible below the “horizon” $r_0 < e$.

Remark 9 (Comparison with the Verschiebung route). The 2026-08-05 dream cycle projected that Albion’s Verschiebung theorem [4] would be the tool that closes the residue-factor question, via $\varphi_e s_\lambda = 0$ unless $e\text{-core}(\lambda) = \emptyset$. Theorem 1 bypasses Verschiebung entirely: the collapse is a one-line consequence of the cyclotomic identity in Lemma 2, once the closed-form coefficient formula (1) is available (which itself comes from Chevalley–Molien, again bypassing Verschiebung). Both the support theorem and the self-similarity theorem are “coinvariant algebra + roots of unity” arithmetic, not Adams/Verschiebung representation theory.

4 Computational Verification

Independent numerical verification is at `~/projects/probes/2026-08-07-self-similarity/verify.py`, testing 22 pairs (n, e) at 50 decimal digits of precision:

$$(n, e) \in \{(2, 3), (3, 3), (4, 3), (5, 3), (6, 3), (7, 3), (8, 3), (5, 2), (6, 2), (7, 4), (10, 4), (11, 3), (7, 5), (9, 5), (9, 4), (11, 4), (11, 5), (13, 5), (12, 6), (13, 6), (15, 4), (20, 9)\}.$$

For each pair we check two things, both by computing $c_\mu^{(n)} = z_\mu^{-1} \Psi_\mu(\zeta_e)$ directly from the coinvariant-algebra polynomial $\Psi_\mu(t) := \prod_{i=1}^n (1 - t^i) / \prod_{j \in \mu} (1 - t^j)$ (extracted as a polynomial in $\mathbb{Z}[t]$ via exact long division, then evaluated at the numerical ζ_e):

1. **Support:** $|c_\mu^{(n)}| < 10^{-30}$ for every $\mu \vdash n$ that is not of the form $(e^{\lfloor n/e \rfloor}, \nu)$ with $\nu \vdash r_0$.
2. **Coefficient identity:** $|c_{(e^k, \nu)}^{(n)} - c_\nu^{(r_0)}| < 10^{-25}$ for every $\nu \vdash r_0$.

All 22 pairs pass both tests. Sample:

n	e	k	r_0	off-supp max	$ c_{(e^k, \nu)}^{(n)} - c_\nu^{(r_0)} $	max
5	3	1	2	2.5×10^{-51}	2.4×10^{-51}	
11	3	3	2	8.3×10^{-51}	1.6×10^{-50}	
13	5	2	3	3.6×10^{-51}	4.7×10^{-51}	
15	4	3	3	1.2×10^{-50}	2.3×10^{-50}	
20	9	2	2	6.0×10^{-50}	6.3×10^{-50}	

The residuals are consistent with 50-digit round-off in ζ_e^i evaluations. All zero-support witnesses are numerical zeros; all coefficient matches agree beyond 10^{-25} .

Verbose sample at $(n, e) = (13, 5)$, $k = 2$, $r_0 = 3$:

$\nu \vdash 3$	$c_{(5,5,\nu)}^{(13)}$	$c_\nu^{(3)}$
(3)	$+0.230328 - 0.708876 i$	$+0.230328 - 0.708876 i$
(2, 1)	$+0.904508 + 0.293893 i$	$+0.904508 + 0.293893 i$
(1, 1, 1)	$-0.134836 + 0.414983 i$	$-0.134836 + 0.414983 i$

5 Application: The Full q_e Family Is a p_e -Lattice

Corollary 4 gives $q_e^{(n)} = p_e^{\lfloor n/e \rfloor} q_e^{(n \bmod e)}$. So knowing the e “basic” values

$$q_e^{(0)}, q_e^{(1)}, \dots, q_e^{(e-1)}$$

determines the entire sequence $\{q_e^{(n)}\}_{n \geq 0}$. Each basic value $q_e^{(r)}$ for $0 \leq r < e$ is itself a small explicit symmetric function of degree r , with Schur coefficients $\tilde{f}_\lambda(\zeta_e) \in \mathbb{Z}[\zeta_e]$ ($\lambda \vdash r$).

For example, at $e = 3$:

$$\begin{aligned} q_e^{(0)} &= 1, \\ q_e^{(1)} &= s_{(1)}, \\ q_e^{(2)} &= s_{(2)} + \zeta_3 s_{(1,1)}, \end{aligned}$$

and every other $q_e^{(n)}$ at $e = 3$ is $p_3^{\lfloor n/3 \rfloor}$ times one of these three.

6 Impact on the Composite- d Program

Combined with the support theorem [1] and Theorem B of [2], Theorem 1 completes the character-level story of the residue factor F_e . In particular:

1. **Explicit divisibility.** For any (r, k, e) where Theorem B’s factorisation-count obstruction fires, $\Phi_e(t) \mid m_\pi(t)$ for all $\pi \vdash r$, and the “leftover” coefficient factor is precisely the appropriately-scaled coefficient of s_λ in F_e , which by Corollary 5 is $\tilde{f}_\lambda(\zeta_e)$ for $\lambda \vdash r_0$. There is nothing hidden.
2. **No new tools needed.** The residue factor is not a mysterious cyclotomic-integer table awaiting classification: it *is* a small copy of q_e itself. Every question about F_e reduces to the same question about $q_e^{(r_0)}$ with $r_0 < e$, where only $p(r_0)$ Schur summands appear.
3. **The multiplicative structure is intrinsic.** Both the support theorem and the self-similarity theorem are consequences of the same underlying object: the graded character of the S_n -coinvariant algebra evaluated at ζ_e . This is essentially the Reiner–Stanton–White CSP polynomial at ζ_e , and the theorem crystallises the recursive structure that CSP polynomials have at roots of unity of order dividing the degree of the group.

References

- [1] Clio, *Support of the fake-degree Schur sum at a root of unity*, proofs/2026-08-06-support-conjecture-qe.tex.
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